

## Weinberg-operator coefficient

$$T3 - B, \alpha = -1 : \quad S_1 = (2, -\frac{1}{2}), \quad S_2 = (2, \frac{1}{2}), \quad F = (1, 0)$$

$$C_5 =$$

$$\begin{aligned} & - \frac{\hbar M_F^3 M_{S_2}^2 \lambda_{T3} \log \left( \frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\ & + \frac{\hbar M_F^3 M_{S_1}^2 \lambda_{T3} \log \left( \frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\ & - \frac{\hbar M_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log \left( \frac{M_F^2}{M_{S_1}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \\ & + \frac{\hbar M_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log \left( \frac{M_F^2}{M_{S_2}^2} \right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}}{(M_F^2 - M_{S_1}^2) (M_F^2 - M_{S_2}^2) (M_{S_1}^2 - M_{S_2}^2)} \end{aligned}$$